

Let C_0 be a baseline copula (empirical or parametric) calibrated from longitudinal data, inducing a transition kernel $F_0(v | u) = \Pr(V \leq v | U = u) = \partial C_0(u, v) / \partial u$.

We posit a *growth regime* $\mathbf{H}_\theta$ for the latent conditional percentile $P \in (0, 1)$ (SGPc scale), with the key assumption $P \perp U$ within the subgroup.

Then the implied current-grade marginal is computed deterministically by

$$F_\theta(v) = \mathbb{E} [H_\theta(F_0(v | U))] \approx \frac{\sum_i w_i H_\theta(F_0(v | u_i))}{\sum_i w_i}.$$

Uniform growth corresponds to $\mathbf{H}(\mathbf{p}) = \mathbf{p}$. Departures from uniformity tilt mass within the baseline kernel and change the implied $\mathbf{V}$ marginal.